Finite volumes on polyhedral cells: exact moments, polynomial reconstruction and the shallow-water equations

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Abstract

We describe the finite-volume method of the Fortran code POLYEDER as it is carried over to the Julia package `dmplex`: cell complexes of general polyhedra with vertex coordinates, the exact integration of monomials over cells and faces, the geometry derived from these moments, a polynomial reconstruction that preserves the cell averages, face gradients from interaction regions around the vertices, and the rotating shallow-water equations on a single layer of prisms over the sphere. The moments are computed by homogeneous numerical integration: Euler's identity and the divergence theorem in the affine hull of each cell, face and edge reduce the integral of a monomial to the vertices, following the cone hierarchy of the cell complex, without the degenerate cases of vertex formulas and for non-convex cells. Working draft.

Keywords: finite volumes, polyhedral meshes, integration of polynomials, k-exact reconstruction, multipoint flux approximation, shallow-water equations

1. Introduction

POLYEDER is a finite-volume code for general polyhedral cells: cut cells over orography, Voronoi and cubed-sphere meshes of a spherical shell, refined and coarsened meshes. Scalars are cell averages, the momentum is the normal component at the faces. Three ingredients carry the method:

1. the polynomial reconstruction of cell averages, for the upwind fluxes of the transport of scalars and momentum, which needs the moments $\int_C (\mathbf{x} - \mathbf{x}_0)^\alpha$ of the cells of a stencil;

2. the gradient and the interpolation to the faces from local problems in the interaction region around each vertex (the corners of the cells at the vertex), a multipoint flux approximation;
3. the velocity at the cells from the normal components at the faces and the projection of the cell tendencies back to the faces.

The shallow-water equations are solved on one layer of prisms over the sphere with free-slip top and bottom, the same code as for the three-dimensional Euler equations.

In `dmplex` the method works on a cell complex (Section 2) that is built from an extruded surface mesh and later from general three-dimensional meshes. The moments are computed by homogeneous numerical integration [6, 3] (Section 3), the geometry of cells and faces from the moments of degree one (Section 4).

2. Cell complexes

A three-dimensional cell complex consists of cells, faces, edges and vertices. The *cone* of a cell is the list of its faces, that of a face the list of its edges, that of an edge its two vertices, each with an incidence number ± 1 of the cellular boundary,

$$\partial C = \sum_{F \in \mathrm{cone}(C)} s_{CF}\, F, \qquad \partial F = \sum_{E \in \mathrm{cone}(F)} s_{FE}\, E, \qquad \partial E = v_2 - v_1, \tag{1}$$

such that $\partial\partial = 0$. The orientation of a face is the circulation of its oriented boundary, its normal follows by the right-hand rule, and $s_{CF} = +1$ when the normal points out of C. A face between two cells has opposite signs in their cones, a face on the boundary of the domain one sign.

`dmplex` stores a surface mesh as a `Plex` (cells, edges and vertices with cones and supports, the analogue of PETSc's `DMPlex`). Its extrusion `ExtrudedPlex` into n layers is the complex of the tensor prisms: a base cell times a layer is a cell, times a level a horizontal face; a base edge times a layer a lateral face, times a level a horizontal edge; a base vertex times a layer a vertical edge, times a level a vertex. Only the surface mesh and n are stored; cones and signs follow from those of the base, $\partial(p \times I) = \partial p \times I + (-1)^{\dim p}\, p \times \partial I$. The `PolyComplex` holds the cones of the three dimensions in compressed form with the vertex coordinates; `extruded_complex` places the vertex v of the base at the level of height z_k at $(a + z_k)\, v$ on a sphere of radius a and at $v + z_k \hat{\mathbf{z}}$ on a plane. A general three-dimensional mesh fills the same arrays.

3. Exact moments of polytopes

3.1. Homogeneous integration

Let $P \subset \mathbb{R}^3$ be a k-dimensional polytope ($k = 1, 2, 3$: a segment, a planar polygon, a polyhedron), $H = \operatorname{aff} P$ its affine hull and f a homogeneous polynomial of degree q, so that $\mathbf{y} \cdot \nabla f = qf$ (Euler). For a point $\mathbf{y}_0 \in H$ the vector field $X = \mathbf{y} - \mathbf{y}_0$ is tangent to H with $\operatorname{div}_H X = k$, and the divergence theorem in H gives

$$\int_P \operatorname{div}_H(Xf) = \int_P \Big(kf + (\mathbf{y}-\mathbf{y}_0)\cdot\nabla f\Big) = (k+q)\int_P f - \int_P \mathbf{y}_0 \cdot \nabla f = \sum_G d_G \int_G f, \tag{2}$$

where G runs over the facets of P with the outward unit normals ν_G in H, and $d_G = (\mathbf{y} - \mathbf{y}_0) \cdot \nu_G$ is constant on G (the distance of $\operatorname{aff} G$ from $\mathbf{y}_0$). For the monomials $\mathbf{y}^\alpha$, $|\alpha| = q$, this is the recursion

$$\boxed{(k + |\alpha|) \int_P \mathbf{y}^\alpha = \sum_G d_G \int_G \mathbf{y}^\alpha + \sum_{i=1}^{3} y_{0,i}\, \alpha_i \int_P \mathbf{y}^{\alpha - e_i}} \tag{3}$$

in the degree for P and down to the vertices in the dimension. With $\mathbf{y}_0$ the foot of the origin on H the last term vanishes for $k = 3$ ($\mathbf{y}_0 = 0$), and on a segment $[\mathbf{a}, \mathbf{b}]$ with the unit tangent $\mathbf{t}$ the facets are its end points with $d_\mathbf{b} = \mathbf{t} \cdot \mathbf{b}$, $d_\mathbf{a} = -\mathbf{t} \cdot \mathbf{a}$ and $\int_{\{\mathbf{v}\}} \mathbf{y}^\alpha = \mathbf{v}^\alpha$. The order-zero moments are the measures: $k\,|P| = \sum_G d_G |G|$, the formula of the volume of a pyramid. For a polygon with the vertices $\mathbf{y}_1, \ldots, \mathbf{y}_m$ in the order of its orientation, the normal $\mathbf{n} = \mathbf{A}/|\mathbf{A}|$ follows from the area vector $\mathbf{A} = \frac{1}{2}\sum_i \mathbf{y}_i \times \mathbf{y}_{i+1}$ (Newell), and the outward normal of the edge from $\mathbf{y}_i$ to $\mathbf{y}_{i+1}$ in its plane is $\nu = \mathbf{t} \times \mathbf{n}$.

Remark 1. POLYEDER integrates the ridge monomials $(\mathbf{b} \cdot (\mathbf{x} - \mathbf{x}_0))^n$ over a cell by a vertex formula with weights of the vertices that depend on $\mathbf{b}$; faces and edges normal to $\mathbf{b}$ need separate cases (`SwitchInt`). The recursion (3) has no such cases, holds for non-convex cells and yields all monomials $\mathbf{y}^\alpha$ of a degree at once; a ridge monomial is a linear combination of them.

3.2. Frames and non-planar faces

The moments of a cell are taken in the local coordinates $\mathbf{y} = (\mathbf{x} - \mathbf{x}_C)/h_C$ with $\mathbf{x}_C$ the mean of its vertices and h_C the largest distance of a vertex from it, so that $|\mathbf{y}| \le 1$ on the cell and the recursion is well conditioned for any position and size of the cell: the returned moments are $\int_C \Big((\mathbf{x} - \mathbf{x}_C)/h_C\Big)^\alpha dV$.

Faces of the cell are integrated in the frame of the cell. A face whose vertices do not lie in one plane — a spherical polygon with more than three vertices — is replaced by the fan of triangles from the mean of its vertices, oriented as the face; both cells of the face see the same triangles, so the complex stays closed. The lateral faces of the prisms are planar, since both of their vertical edges lie on radii.

3.3. Tests

The moments of all monomials up to degree four are exact to round-off on the unit cube, on a triangular prism and on a non-convex L-shaped prism (`test/polyhedral.jl`), as is the divergence theorem $\int_C \partial_i \mathbf{y}^\alpha = \sum_F s_{CF}\, n_{F,i} \int_F \mathbf{y}^\alpha$ with the face moments.

4. Geometry

The volume and the centroid of a cell are its moments of degree zero and one, the area vector of a face is the sum of those of its planar pieces, its centroid follows from the face moments. The outward area vectors of every cell sum to zero (to 10^{-17}, Table 1), which makes the divergence of a constant field vanish exactly. On a spherical shell of thickness d the prisms have flat faces inscribed in the shell; their total volume converges at second order to that of the shell, on triangles, hexagons and the squares of the cubed sphere.

Table 1: Single layer of prisms on the unit sphere, thickness 0.1: relative error of the total volume against $\frac{4\pi}{3}(1.1^3 - 1)$ and the largest $|\sum_F s_{CF}\mathbf{A}_F|$ of a cell.

mesh	cells	volume error	closure
icosahedral, level 2	320	-3.38×10^{-2}	2×10^{-17}
icosahedral, level 3	1280	-8.61×10^{-3}	1×10^{-17}
icosahedral, level 4	5120	-2.16×10^{-3}	5×10^{-18}
hexagonal, level 2	162	-2.66×10^{-2}	3×10^{-17}
hexagonal, level 3	642	-6.76×10^{-3}	1×10^{-17}
hexagonal, level 4	2562	-1.70×10^{-3}	5×10^{-18}
cubed sphere, 4×4	96	-6.51×10^{-2}	6×10^{-17}
cubed sphere, 8×8	384	-1.67×10^{-2}	2×10^{-17}
cubed sphere, 16×16	1536	-4.21×10^{-3}	9×10^{-18}

5. Polynomial reconstruction

5.1. Average-preserving least squares

For every cell C we seek the polynomial $p_C = \sum_k c_k \, \mathbf{y}^{\alpha_k}$ of degree $\le r$ in the local coordinates $\mathbf{y} = Q_C(\mathbf{x} - \mathbf{x}_C)/h_C$ ($\mathbf{x}_C$ the centroid, h_C the largest distance of a vertex, Q_C orthogonal) that has the average $\bar{u}_C$ of the cell and fits the averages of the other cells of its stencil S_C in the least-squares sense [2, 7],

$$\frac{1}{|C|}\int_C p_C = \bar{u}_C, \qquad \sum_{C' \in S_C \setminus C} \Big(\frac{1}{|C'|}\int_{C'} p_C - \bar{u}_{C'} \Big)^2 \to \min . \tag{4}$$

With $a_{C'k} = |C'|^{-1} \int_{C'} \mathbf{y}^{\alpha_k}$ (the moments of Section 3 in the frame of C) the constraint eliminates the constant, $c_0 = \bar{u}_C - \sum_{k\ge1} a_{Ck} c_k$, and the remaining coefficients solve

$$\sum_{k\ge1} (a_{C'k} - a_{Ck})\, c_k = \bar{u}_{C'} - \bar{u}_C, \qquad C' \in S_C \setminus C, \tag{5}$$

by the pseudo-inverse with a relative rank tolerance (POLYEDER's `TolRank`). The coefficients are linear in the averages of the stencil, $c = R_C \, \bar{u}_{S_C}$, computed once.

Stencils. S_C is C with the cells that share a vertex with it, grown ring by ring until the reduced matrix has full rank and at least 1.5 times as many rows as monomials. Inside a mesh two rings suffice for degree three; at boundaries the stencil grows, for instance in the bottom and top layers of a column, where two rings see only three layers.

Tangential basis. On one layer of cells over the sphere the stencil does not resolve the radial structure, and the radial monomials make the least-squares problem singular. The option `basis = :tangential` takes Q_C with the radius at the centroid as its third axis and uses only the monomials without the third coordinate: a reconstruction in the tangent plane, the moments still exact over the prisms.

Face values. The average of p_C over a face F of C is $|F|^{-1} \int_F p_C = \sum_k c_k \, |F|^{-1} \int_F \mathbf{y}^{\alpha_k}$; for the two cells of every face these are two sparse matrices from the cell averages to the face values (`face_value_matrices`), and at the points of a face quadrature two more (`point_value_matrices`). On a distorted triangle mesh of a box, extruded into five layers, all face values of polynomials of degree ≤ 3 are reproduced to 10^{-12}.

5.2. Transport

The transport $\partial_t q + \nabla \cdot (q\mathbf{u}) = 0$ of the cell averages takes the upwind values at the faces. With one volume flux $U_F = \int_F \mathbf{u} \cdot \mathbf{n}$ per face, as for the normal velocities of a staggered scheme, the flux is $U_F\, \bar{q}_F$ with the upwind face average. This is only second order, whatever the degree: $\int_F q\, \mathbf{u} \cdot \mathbf{n} - U_F \bar{q}_F = O(h^2)\, |F|$ unless $\mathbf{u} \cdot \mathbf{n}$ is constant on F. With the velocity field known at the points $\mathbf{x}_g$ of a face quadrature (`face_quadrature`: the seven-point rule of degree five [4] on the fan of triangles of the face) the flux $\sum_g w_g (\mathbf{u} \cdot \mathbf{n})_g\, q_g$, upwinded point by point, has the order $r + 1$ of the reconstruction. Time integration is the third-order SSP Runge–Kutta method.

Test fluxes that are exactly divergence free in every cell follow from a vector potential: for the velocity $\mathbf{u} = g(\eta) \nabla\psi \times \nabla\eta$ of a stream function ψ of the horizontal position (η the radius on the sphere, the height z on a plane) the flux through F is the circulation of $g(\eta)\psi\nabla\eta$ around F, $U_F = \sum_{a \to b} \frac{1}{2}(\psi(a) + \psi(b))\, (G(\eta_b) - G(\eta_a))$ over the oriented edges of F with $G' = g$; it vanishes on the faces of constant η, and the fluxes of a cell sum to zero exactly. On the sphere $\psi = \boldsymbol{\Omega} \cdot \hat{\mathbf{r}}$ with $G(\eta) = \eta^3/3$ gives the solid-body rotation $\boldsymbol{\Omega} \times \mathbf{x}$.

Table 2: Steady flow $\mathbf{u} = (\sin \pi x \cos \pi y, -\cos \pi x \sin \pi y)$ in the unit square, distorted triangles ($2n^2$ cells, one layer), $q = \exp(2\psi)$, $\psi = \sin \pi x \sin \pi y / \pi$, relative L^2 error at $t = 0.5$, fluxes at the quadrature points; CFL 0.4.

	degree 1		degree 2		degree 3	
n	error	order	error	order	error	order
8	9.57×10^{-3}		4.98×10^{-3}		2.71×10^{-3}	
16	2.09×10^{-3}	2.20	6.97×10^{-4}	2.84	2.16×10^{-4}	3.65
32	4.21×10^{-4}	2.31	9.69×10^{-5}	2.85	1.05×10^{-5}	4.37
64	9.76×10^{-5}	2.11	1.27×10^{-5}	2.93	7.30×10^{-7}	3.84

With the face fluxes $U_F \bar{q}_F$ the same flow converges at second order for all degrees (1.2×10^{-4}, 1.0×10^{-4}, 1.3×10^{-4} at $n = 64$). On one layer over the sphere (thickness 0.01) a Gaussian hill $\exp(-|\hat{\mathbf{r}} - \hat{\mathbf{r}}_0|^2/0.4^2)$ is rotated once about an axis tilted by 45° with the divergence-free face fluxes (Table 3); mass is conserved to 10^{-14}. Besides the flux, the flat faces of the prisms inscribed in the shell limit the order to two.

Remark 2. For the shallow-water equations with the normal momentum at the faces, the mass flux through a face is a single number, and the transport

Table 3: Solid-body rotation of a Gaussian hill over one revolution on one layer of prisms over the unit sphere, tangential basis, face fluxes $U_F \bar{q}_F$: relative L^2 error.

mesh	cells	degree 0	degree 1	degree 2	degree 3
hexagonal 3	642	7.6×10^{-1}	3.3×10^{-1}	3.7×10^{-1}	1.3×10^{-1}
hexagonal 4	2562	6.3×10^{-1}	1.2×10^{-1}	1.2×10^{-1}	2.2×10^{-2}
hexagonal 5	10242	4.7×10^{-1}	3.3×10^{-2}	2.2×10^{-2}	4.5×10^{-3}
icosahedral 3	1280	6.1×10^{-1}	2.3×10^{-1}	1.4×10^{-1}	9.8×10^{-2}
icosahedral 4	5120	4.6×10^{-1}	7.0×10^{-2}	3.1×10^{-2}	2.0×10^{-2}
icosahedral 5	20480	3.1×10^{-1}	1.8×10^{-2}	5.7×10^{-3}	4.6×10^{-3}

is second order like that of every staggered scheme; the higher reconstruction lowers the error constant and the dispersion (Table 3). Higher order needs the normal velocity along the face (from the reconstructed cell velocity, or several unknowns per face) and cells with curved faces on the sphere.

6. Interaction regions and the face gradient

Around every vertex v the cells at v (the *corners*) and the faces at v form the interaction region of a multipoint flux approximation [1] (POLYEDER: `TGrad`). Every corner has a linear gradient from the values at the centroids $\mathbf{x}_F$ of its faces at v,

$$\mathbf{g}_C = G_C \, (u_F - \bar{u}_C)_{F \ni v}, \qquad G_C = \left[(\mathbf{x}_F - \mathbf{x}_C)^T \right]^+_F, \tag{6}$$

three faces for the corners of prisms (the pseudo-inverse for more). The values u_F at the faces of the region are the unknowns of a local linear system: the normal gradient is continuous, $\mathbf{n}_F \cdot \mathbf{g}_{C_1} = \mathbf{n}_F \cdot \mathbf{g}_{C_2}$, across every interior face, and vanishes on the boundary. Its solution expresses the half-face values and normal gradients in the cell averages of the region; the face value and the normal gradient at a face are their means over its vertices, weighted by the share of the vertex in the area of the face (the quadrilateral of the vertex, the midpoints of its edges and the mean of the vertices). The two sparse matrices are exact for linear fields at all faces whose vertices lie inside the domain (to 10^{-14} on distorted triangular prisms in four layers). On one layer over the sphere the zero normal gradient at the top and the bottom is exact for every function of the horizontal position.

7. The shallow-water equations

On one layer of prisms over the sphere, free slip at the top and the bottom, the height h is a cell average and the normal momentum $U_F = (h\mathbf{u})\cdot\mathbf{n}_F$ lives on the lateral faces:

$$\frac{d\bar{h}_C}{dt} = -\frac{1}{|C|}\sum_F s_{CF}|F|\,U_F, \tag{7}$$

$$\frac{dU_F}{dt} = \mathbf{n}_F\cdot\Big[\mathbf{T}_C\Big]_F - \partial_n\Big(\tfrac{g}{2}h^2\Big)_F - g\,h_F\,\partial_n b_F, \qquad \mathbf{T}_C = -\frac{1}{|C|}\sum_F s_{CF}|F|\,\frac{U_F}{h_F}\,(\mathbf{m})_F^{\mathrm{up}} - f\,\hat{\mathbf{r}}\times\mathbf{m}_C. \tag{8}$$

The pressure gradient, the bathymetry and the face values h_F come from the interaction regions; the cell momentum $\mathbf{m}_C$ is reconstructed from the normal components, its Cartesian components are transported with the upwind values of the polynomial reconstruction (Section 5, tangential basis), and the cell tendency $\mathbf{T}_C$ is carried to the faces with the face values of the interaction regions, again exact for linear fields. Mass is conserved to round-off. Time integration: the third-order SSP Runge–Kutta method at the gravity-wave CFL number 0.4.

The cell momentum. Perot's reconstruction [8] (POLYEDER's `SumNormal`),

$$\mathbf{m}_C = \frac{1}{|C|}\sum_F s_{CF}|F|\,U_F\,(\mathbf{x}_F - \mathbf{x}_C), \tag{9}$$

is exact for constant fields; for linear fields the face-centroid rule misses a quadratic term, an $O(h)$ error that cancels on symmetric cells only. On triangles the scheme is therefore first order, and no reconstruction from the three normal components of the cell alone can do better: they define exactly one lowest-order Raviart–Thomas field $\mathbf{a} + b(\mathbf{x} - \mathbf{x}_C)$, three degrees of freedom against the six of a linear tangential field, and $\mathbf{m}_C$ is its value at the centroid. The barycentric reconstruction (POLYEDER's `Barycentric`: at every corner the vector with the normal components of its three faces, the cell vector their Wachspress mean) gives the same: the corner vectors are the values of that field at the vertices. Its matrix agrees with Perot's to 10^{-7} (the curvature of the layer), and on hexagons it is about twice less accurate than Perot's. A least-squares fit of a linear vector field to the normal components on the faces of the cell and of its vertex neighbours is exact for linear

fields, but unstable in the shallow-water scheme; weighting the faces of the cell itself twenty times stronger than those of the neighbours makes it stable and second order on triangles. On hexagons and quadrilaterals Perot's form is more accurate and stays stable over 15 days, where the weighted fit grows; the default takes the fit on triangles only. The instability of the unweighted fit is not one of the energy: projecting the cell tendency with the adjoint of the reconstruction R, $P = (R^\top V R)^{-1} R^\top V$, makes the Coriolis term free of work but blows up within 20 steps. $R^\top V R$ is singular on hexagons (three faces per cell against two components) and badly conditioned on triangles, the ratio of its extreme eigenvalues being 6.9×10^{-5} for the unweighted fit, 1.5×10^{-3} with the weight and 3.7×10^{-3} for Perot's: the fit hardly sees some face patterns, which then grow unchecked. The weight converges for large values (2.094×10^{-3} and 5.66×10^{-4} on levels 3 and 4 for weights 10^2 and 10^4), towards the fit constrained to the three normal components of the cell.

Implementation. The operators are sparse matrices of the global complex: the face gradient and the face values of the interaction regions, the upwind face values of the reconstruction from either side and the cell momentum. On every rank they are restricted to padded tables of the rows it computes, in local numbering, and the tendency is four KernelAbstractions kernels gathering over them, without atomics: the cell momentum at the owned cells, the face values and fluxes at the interior faces of the owned cells, the cell tendencies, and the projection with the pressure gradient at the owned faces. With MPI the cells are partitioned into the zonal equal-area regions, a face is owned by the smallest part of its cells, and the ghosts of a rank are exactly the cells and faces in the stencils of its rows; the state, the cell momentum and the cell tendency are exchanged once each. The distributed state equals the serial one bitwise. On the level-6 triangles (81920 cells) a tendency takes 5.7 ms on one core and 3.0 ms on four, against 29 ms for the sparse-matrix form; the same kernels run on GPUs.

Galewsky's jet. At 50–60 km (icosahedral and hexagonal level 7, cubed sphere 200^2; one A100, 6 days in 42–143 s) the scheme stays stable and conserves mass to 10^{-12}, but the grid excites the barotropic instability: the unperturbed jet, balanced within 8–20 m after 3 days, has broken up by day 6 with the wave number of the grid (5 on hexagons, 4 on the cubed sphere, 6 for the fit on triangles). With the perturbation the hexagons and the cubed sphere

Table 4: Williamson's case 2 after 5 days on one layer of prisms (thickness 1000 m), reconstruction of degree 2: relative L^2 error of h; mass conserved to 10^{-13}.

mesh	cells	Perot	weighted fit	default
icosahedral 3	1280	3.96×10^{-3}	2.10×10^{-3}	fit
icosahedral 4	5120	2.00×10^{-3}	5.68×10^{-4}	
icosahedral 5	20480	1.04×10^{-3}	1.43×10^{-4}	
hexagonal 3	642	2.74×10^{-3}	3.47×10^{-3}	Perot
hexagonal 4	2562	6.69×10^{-4}	1.09×10^{-3}	
hexagonal 5	10242	1.73×10^{-4}	2.96×10^{-4}	
cubed sphere 8^2	384	6.63×10^{-3}	1.03×10^{-2}	Perot
cubed sphere 16^2	1536	7.67×10^{-4}	1.22×10^{-3}	
cubed sphere 32^2	6144	1.84×10^{-4}	2.85×10^{-4}	

show the expected roll-up behind it on top of the grid waves. Perot's vectors on triangles (300 m off balance after 3 days) are too diffusive, and there the perturbation makes no difference.

The terms one by one. The momentum advection alone ($g = 0$, $\Omega = 0$) for a smooth tangential field and height, against $-\mathbf{n}_F \cdot \nabla_s \cdot (h\mathbf{u} \otimes \mathbf{u})$ at the face centroids, splits into the cell momentum, its transport and the projection (Table 5). The transport with the exact cell momentum and the projection of the exact cell tendency converge at first and second order in the maximum norm. The error of the cell momentum, however, enters through the divergence of the fluxes: Perot's vectors on triangles (first order everywhere) give an error of 0.12 that does not decrease, and on the hexagons their error is first order within three cells of the pentagons, so that the advection stalls at 6×10^{-3} on level 6. The weighted fit makes the cell momentum second order on all meshes and the advection converges (hexagons 1.5×10^{-3} on level 6). Galewsky's jet does not improve with it (unperturbed hexagons 13 m and 563 m, cubed sphere 16 m and 368 m after 3 and 6 days), so the advection is not what excites the jet. Splitting the residual of Williamson's case 2, which is a steady state, into advection with Coriolis and the pressure term shows the other suspect: the pressure gradient of the interaction regions is first order in the maximum norm on all meshes (relative 9.5×10^{-4}, 1.3×10^{-3} and 5.4×10^{-4} on hexagons 6, triangles 6 and the cubed sphere 64^2, halving with h), while its mean error is second order, and on the triangles it dominates

the residual.

Table 5: Momentum advection, relative maximum errors at the faces: total, cell momentum $\mathbf{m}_C$, cell tendency and face tendency with the exact $\mathbf{m}_C$.

mesh	$\mathbf{m}_C$	level	total	$\mathbf{m}_C$	cell, exact $\mathbf{m}_C$	face, exact $\mathbf{m}_C$
triangles	fit	4	8.6×10^{-3}	2.0×10^{-3}	1.7×10^{-2}	7.1×10^{-3}
		6	1.6×10^{-3}	1.2×10^{-4}	4.3×10^{-3}	1.6×10^{-3}
triangles	Perot	4	1.2×10^{-1}	2.6×10^{-2}		
		6	1.2×10^{-1}	6.6×10^{-3}		
hexagons	Perot	4	2.2×10^{-2}	3.0×10^{-3}	5.8×10^{-3}	1.9×10^{-2}
		6	5.7×10^{-3}	4.6×10^{-4}	4.0×10^{-4}	1.2×10^{-3}
hexagons	fit	6	1.5×10^{-3}	2.0×10^{-4}		
cubed sphere	Perot	16^2	2.6×10^{-2}	3.1×10^{-3}	7.0×10^{-3}	2.2×10^{-2}
		64^2	2.8×10^{-3}	2.0×10^{-4}	1.2×10^{-3}	1.3×10^{-3}

Next: the pressure gradient near the irregular vertices (the grid imprinting of the jet), bathymetry (Williamson's case 5), split-explicit time integration [9], and the three-dimensional equations on cut cells (Section 8).

8. Cut cells

As in POLYEDER's `CutGrid`, orography is represented by cutting the cells of a complex with the surface $\ell(\mathbf{x}) = 0$ of a level function, the fluid where $\ell > 0$ (`cut_complex`): vertices below the surface are removed; an edge between a vertex below and one above is cut at the root of ℓ on it (Illinois method) and keeps its orientation; a face with cut edges is closed by the edge from the cut point where its oriented boundary leaves the fluid to the one where it enters it again; a cut cell is closed by the terrain face through the closing edges of its faces, each with the sign opposite to that of its face in the cell, so that the boundary of the cell has no boundary. The terrain face is in general not planar and is integrated as the fan of its vertex mean (Section 3). A face crossed twice by the surface is rejected, as in POLYEDER.

Cutting leaves cells of any size: on jittered triangular prisms over $[-1,1]\times[0,\frac{1}{4}]\times[0,1]$ ($2n_x{\cdot}2{\cdot}n_z$ cells) under the mountain $z_s = 0.03{+}0.25\,e^{-(x/0.4)^2}\cos^2(\pi x/0.5)$ the smallest cut cell has $2.5{\times}10^{-11}$ of the volume of its prism. `merge_small_cells` unites every cell below half the volume of its uncut cell with the neighbour across its largest face, repeated for unions still too small; the union of cells

is again a polyhedron of the complex (`merge_cells`: the faces between them removed), with its exact moments, so nothing else changes. The test flow $\mathbf{u} = \nabla \times (\psi\,\hat{\mathbf{y}})$, $\psi = (z - z_s(x))(1 - z)\sin\frac{\pi(x+1)}{2}$, follows the terrain and vanishes on the whole boundary; its fluxes (`vector_potential_flux`, the circulation of $\psi\hat{\mathbf{y}}$ by the trapezoidal rule) are divergence free in every cell and zero through the terrain faces, whose vertices lie on $\psi = 0$. Every cell closes ($|\sum_F s|F|\mathbf{n}_F| < 10^{-17}$), the volume converges at second order to the exact one, mass is conserved to 10^{-15}, and $q = e^{2\psi}$ is steady (Table 6).

The quadratic reconstruction needs more cells at the terrain: with the default stencil (1.5 times as many cells as monomials) a cut cell has a one-sided stencil of 16 cells, and the semi-discrete operator an eigenvalue with $\Re\lambda\,\Delta t = 0.5$, which blows up; twice as many cells still give 0.27, two and a half times as many (`boundary_oversampling` $= 2.5$, at cells with a boundary face) leave 10^{-6} to 10^{-3}. Degrees 0 and 1 are stable with the default.

Table 6: Cut prisms under the mountain, steady transport of $q = e^{2\psi}$ to $t = 0.5$, degree 2, boundary oversampling 2.5: relative L^2 error and SSP-RK3 steps at CFL 0.4 of the upwind limit, without and with merging the cells below half their prism.

$n_x \times n_z$	cut cells	smallest	error	steps	error merged	steps
16×8	82	2.5×10^{-11}	2.01×10^{-2}	74638	2.18×10^{-2}	26
32×16	180	2.1×10^{-6}	6.08×10^{-3}	3150	6.43×10^{-3}	64
64×32	368	3.0×10^{-5}	1.69×10^{-3}	2378	1.72×10^{-3}	119

Next: the interaction regions and the momentum on cut cells, the three-dimensional Euler equations over orography, cut cells on the sphere (with the radial interpolation of POLYEDER's edge search).

Code availability

`src/Polyhedral/` of `dmplex`: `Complex.jl` (`PolyComplex`, `extruded_complex`), `Integration.jl` (`MonomialBasis`, `segment_moments`, `polygon_moments`, `face_moments`, `cell_moments`), `Geometry.jl` (`PolyGeometry`), `Reconstruction.jl` (`PolyReconstruction`, `face_value_matrices`, `point_value_matrices`), `Advection.jl` (`PolyAdvection`, `stream_function_flux`, `vector_potential_flux`), `InteractionRegion.jl` (`interaction_operators`, `cell_vectors`, `linear_vector_reconstruction`), `ShallowWater.jl` (`PolyShallowWater`), `CutCell.jl` (`cut_complex`, `merge_cells`, `merge_small_cells`); tests in `test/polyhedral.jl`.

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